

Fed-IoMT-Block: A Privacy-Preserving Framework for Secure Federated Learning in Consumer-Centric Internet of Medical Things

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Abstract—The rapid proliferation of the Internet of Medical Things (IoMT) has enabled sophisticated, timely, and remote healthcare monitoring by leveraging advanced networking technologies. However, despite ongoing research, the IoMT paradigm still faces significant security and privacy challenges. To address these, this study introduces Fed-IoMT-Block, an end-to-end framework for securing IoMT devices. The framework first authenticates IoMT entities using the Quantum Authentication Protocol (QAP), which identifies communicating entities and secures their exchange using unique quantum particles, providing stronger protection than conventional protocols. Once authenticated, entities share data through federated learning, leveraging the Attention Capsule Network (Att-CapsNet) and a homomorphic algorithm. Network monitoring is then handled by a machine learning agent, the Vulnerability Analysis Agent (VAA), which inspects external traffic and classifies it using the Light Gradient Boosting Machine (LGBM). The associated risk is mathematically quantified based on multiple parameters. Finally, role-based access control is enforced via a Flexible Neuro-Fuzzy Inference System (FN FIS) to prevent unauthorized access. The proposed framework demonstrates notable improvements in packet delivery, security, and scalability compared to existing methods. These improvements include a 26% reduction in malicious traffic, 84.88% accuracy in disease severity detection, 12% lower latency, and a high packet throughput ratio. This study highlights the potential of federated learning and blockchain to

create secure, scalable, and privacy-preserving IoMT systems for consumer healthcare.

Index Terms—Blockchain, federated learning, homomorphic encryption, Internet of Medical Things (IoMT), intrusion detection systems (IDS), quantum authentication, risk assessment.

I. INTRODUCTION

THE INTERNET of Things (IoT) is an emerging technology used for real-time monitoring, data transmission, and various applications across sectors such as emergency services, traffic congestion, industrial control, logistics, retail, and smart cities [1], [2], [3]. Due to the significance of IoT, its feasibility in medical applications for health monitoring has led to a new intelligent paradigm in Healthcare 4.0, referred to as the Internet of Medical Things (IoMT) [4], [5], [6], [7]. IoMT utilizes IoT-based smart medical devices or sensors for continuous health monitoring of patients across various rankings and gradings.

The patient health monitoring process in IoMT typically consists of three methods [8]: (1) On-body medical sensors, deployed on the patient's body to monitor parameters such as heart rate, respiratory rate, and temperature. (2) In-body sensors, installed inside the body to monitor internal organs such as the liver, kidneys, and heart using wireless endoscopy sensors. (3) Off-body sensors, placed around the patient to monitor physical activities such as postural movements or improper seating positions on a sofa. Due to these diverse monitoring applications, the proportion of IoT devices is projected to increase by 50% by 2030, marking a 150% rise over current levels [2], [9].

The key advantages of IoMT include cost reduction, improved quality of life, and enhanced user experience [10], [11]. However, a major concern is its security and privacy [12]. Communication among IoMT devices occurs in a wireless environment, making them attractive targets for attackers aiming to perform malicious actions that may endanger human lives [13], [14]. Common threats include eavesdropping, intentional delays in device detection capabilities, and Distributed Denial of Service (DDoS) attacks. A DDoS attack is one of the widespread attacks that tends to flood a legitimate server to prevent access by legitimate users by consuming the entire bandwidth. By sending flooding messages to the server, it appears that the server is not showing

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online availability to legitimate users, and in this way, the server's services are hijacked.

According to a report by the International Medical Cybersecurity Council (IMCC), nearly 70% of smart medical devices are highly vulnerable to cyber-attacks due to the lack of effective authorization and authentication mechanisms [15], [16], [17]. Additionally, IoMT systems face challenges related to scalability and storage limitations, underscoring the need for extensive research in IoMT cybersecurity.

Authentication and authorization have been identified as the primary security approaches in the literature to counter cyber threats [18], [19]. Current research largely relies on cryptographic algorithms such as Elliptic Curve Cryptography (ECC) and the Digital Signature Algorithm (DSA). However, these algorithms are often unsuitable for compact IoMT devices due to their limited computational, storage, and processing capacities, which ultimately compromises security [20]. Although technologies such as Artificial Intelligence (AI) and blockchain have shown promise, their application in Intrusion Detection and Prevention Systems (IDS/IPS) for IoMT remains limited due to their high computational and resource demands [21]. Moreover, AI models are susceptible to insider threats, poisoning attacks, and fake malicious traffic.

To address scalability, flexibility, and security concerns, cloud-based on-demand services have been adopted alongside blockchain to ensure data integrity, tamper resistance, confidentiality, and immutability [22], [23]. However, their use in IoMT environments has not been adequately explored for comprehensive end-to-end authentication, authorization, model protection, privacy, and network security. To bridge this gap, we propose an end-to-end security strategy for IoMT applications using Federated Learning (FL) and blockchain technologies.

The major security challenges addressed in state-of-the-art research [2], [4], [7], [10], [13], [15], [22], [23], [24], [25] are outlined below:

- **Inadequate Smart Medical Device Validation:** The authenticity and legitimacy of smart IoMT devices are typically verified using IDs, passwords, and biometrics. However, these methods fail to ensure full legitimacy, leaving devices vulnerable to spoofing attacks.
- **Lack of Risk Assessment:** Security and privacy-preserving strategies are weakened when risk assessment is not performed, negatively affecting Quality of Service (QoS) and network credibility.
- **Significant Security Loopholes:** Most existing work emphasizes securing medical data via encryption and federated learning to guard against external threats, but offers limited analysis of malware targeting smart IoMT devices.

Motivated by the above challenges, the proposed study aims to enhance security and privacy in the IoMT environment by designing a secure and private framework leveraging advanced technologies. The novel contributions of this study are as follows:

- This research study uses a federated machine learning model to develop a novel security and privacy paradigm for the heterogeneous IoMT devices environment by incorporating communication and network security measures including Intrusion Detection Systems (IDS) and risk assessment.

- The adoption of the Quantum Authentication Protocol (QAP), supported by blockchain, enhances IoMT security and privacy beyond conventional cryptographic protocols.
- The integration of AI and homomorphic encryption with federated learning offers additional privacy for data auditing and integrity without requiring decryption or cloud-side access to raw data.
- Performance evaluations show that the proposed Fed-IoMT-Block framework achieves high accuracy in malicious device and attack detection, improved disease severity classification, and reduced latency with higher packet throughput. The framework also effectively defends against Man-in-the-Middle, DDoS, insider, and poisoning attacks.

The remainder of this paper is organized as follows: Section II presents a review of related work. Section III describes the system model and prerequisites. Section IV details the proposed processes with supporting pseudocode, diagrams, and equations. Section V discusses the experimental setup, comparative analysis, and security evaluation. Section VI concludes the paper.

II. LITERATURE REVIEW

This section reviews key shortcomings in state-of-the-art efforts to secure the IoMT environment. It is organized into three subsections to highlight existing gaps in the literature.

A. Blockchain Approaches for IoMT Security

Several studies have explored blockchain technology to enhance IoMT device security. The edge computing and blockchain approach by [26] reduces computation and latency in data transmission using distributed edge servers, IoMT devices, smart contract-based blockchains, and IPFS storage. Data execution and offloading are managed via mobile devices and Patient IDs. However, the involvement of multiple entities with different authentication and authorization processes makes it unsuitable for resource-constrained devices.

The bidirectional Long Short-Term Memory (LSTM) and smart contract-based blockchain method in [27] detects misbehavior in IoMT pancreas devices but does not account for the computational burden on lightweight devices. Reference [28] used blockchain to ensure secure communication and synchronization in critical medical infrastructure. A blockchain-based access control method in [29] stores device data in digital wallets using decentralized identity devices, later analyzed in the fog layer. However, reliance on manual authentication based on credentials, roles, and attributes introduces vulnerabilities.

The blockchain-based Covichain framework in [30] secures IoMT data transfers for COVID-19 patients but incurs high computation, processing, and storage costs due to IPFS. Reference [31] addressed centralized storage limitations by using distributed storage with ring-assisted access control for authenticating patients, doctors, and nurses. However, the study did not address cost concerns in the IoMT setting.

Reference [32] proposed a context-aware attribute-based encryption scheme for secure IoMT data. However, its reliance

on cloud and fog servers, combined with blockchain and biomedical sensors, results in high resource consumption.

B. Federated Learning Approaches for IoMT Security

Federated Learning (FL) has been integrated into healthcare using blockchain, as demonstrated in [33]. A similar approach in [34] incorporated enhanced secret sharing with FL. Here, user authentication is handled via a dynamic hash function, with keys shared using Elliptic Curve Cryptography (ECC)—also applied in [35]. A dual-phase method combining privacy-preserving techniques and FL blocks was proposed to address insider and denial-of-service attacks. However, reliance on centralized servers contradicts the decentralized nature of FL. Another architecture with smart IoT devices, fog servers, and gateways for encryption was presented in [36], demonstrating lower communication costs than SPPDA and EHDA.

Blockchain has also been applied for data preprocessing [37]. A fraud detection model combining FL and blockchain was introduced in [38], incorporating application and fog layers. Reference [39] proposed a masking scheme using homomorphic encryption in FL, applying a weighted average algorithm to boost resiliency. Similarly, [40] employed two FL algorithms, S-LMI and C-GMI, for data auditing and verification to defend against poisoning attacks. Both use homomorphic hash functions to download data from individual IoMT devices. However, the scheme does not quantify cost complexity or preemptively detect such attacks. A fuzzy ensemble-based federated learning model was also introduced using temporal convolutional networks (TCN), Long Short-Term Memory (LSTM), and gated recurrent units (GRU) [41]. While cybersecurity and cost concerns are addressed, a hybrid FL-based quantization framework was later proposed [42], [43], [44] to improve protection against data leakage during transmission, though cost complexity remains unaddressed. However, none of these schemes fully exploit FL due to the high computational, processing, and storage demands of IoMT devices.

C. Other Security Approaches for IoMT Security

The two-phase Secure-IoMT framework using multimodal sensory signals was proposed in [45], comprising pre-processing, clustering, feature extraction, and classification. However, it does not consider cost complexity. An AI-enabled lightweight communication scheme using IPv6, ZigBee, BLE, Z-Wave, and NFC was introduced in [46], but it lacks active misbehavior detection.

A secure data aggregation scheme with an authentication framework was proposed in [47], employing the HMAC protocol for end-to-end authentication and using a fuzzy-based delimiter for better resource use. Fuzzy-based Trust Management (FTM) in [48] enhances device trustworthiness by identifying Sybil attacks and untrusted nodes. Reference [49] applied Bayesian optimization and deep learning to reduce malicious access in IoMT environments.

III. SYSTEM MODEL

This section outlines the proposed IoMT security framework, which integrates blockchain and federated learning technologies, as shown in Figure 1. Federated learning

preserves patient privacy by avoiding disclosure of original identities, while blockchain ensures transparency, integrity, and immutability within the IoMT environment. The framework comprises three layers: IoMT Sensor Layer, Cloud Layer, and Application Layer. The IoMT Sensor Layer includes temperature sensors, heartbeat sensors, smart ECG machines, and biosensors for continuous health monitoring. The Cloud Layer features a Trusted Authority (TA), Vulnerability Analysis Agent (VAA), and Secure Aggregator (SA) to authenticate devices, users, and doctors, and to detect suspicious network activity. The Application Layer consists of end users, doctors, and medical assistants who access patient data through an access control mechanism.

A. Assumptions

The following assumptions are made in the proposed model:

- Malicious insiders: Suspicious digital sources offer various incentives to medical doctors, healthcare assistants, and authenticators (TA, VAA, SA) to update components of the existing system. These updates can be exploited, potentially leading to system breaches caused by insider threats and malicious attacks.
- The TA authenticates IoMT devices, users, and doctors. The VAA monitors suspicious network behavior. The SA manages global model aggregation in FL and handles blockchain transactions and storage.
- Centralized cloud architecture does not hinder scalability, as patient enrollment (i.e., IoMT devices) is limited based on capacity.
- The blockchain is assumed resistant to 51% attacks due to the low number of malicious insiders.

B. Design Goals

The proposed model aims to achieve the following:

- Security and Correctness: A secure TA ensures authenticity without compromising data integrity.
- Scalability: User interaction with the cloud is limited to maintain scalability.
- Privacy and Transparency: Blockchain integration ensures data privacy while supporting transparency.
- Integrity: Federated learning preserves the integrity of sensed data against theft or tampering.

IV. FED-IoMT-BLOCK MODEL

The proposed Fed-IoMT-Block model consists of four sequential steps: (1) Multi-Credential Quantum-based Authenticity Verification, (2) Federated Learning-based Sensed Data Sharing, (3) Machine Learning-Entrenched Attack Detection & Risk Analysis, and (4) Adaptive Fuzzy Role-based Access Control.

A. Multi-Credential Quantum-Based Authenticity Verification

The proposed Quantum Authentication Protocol (QAP) is based on the Greenberger–Horne–Zeilinger (GHZ) quantum state, a fundamental concept in quantum communication theory. The protocol involves three major entities: cloud-based

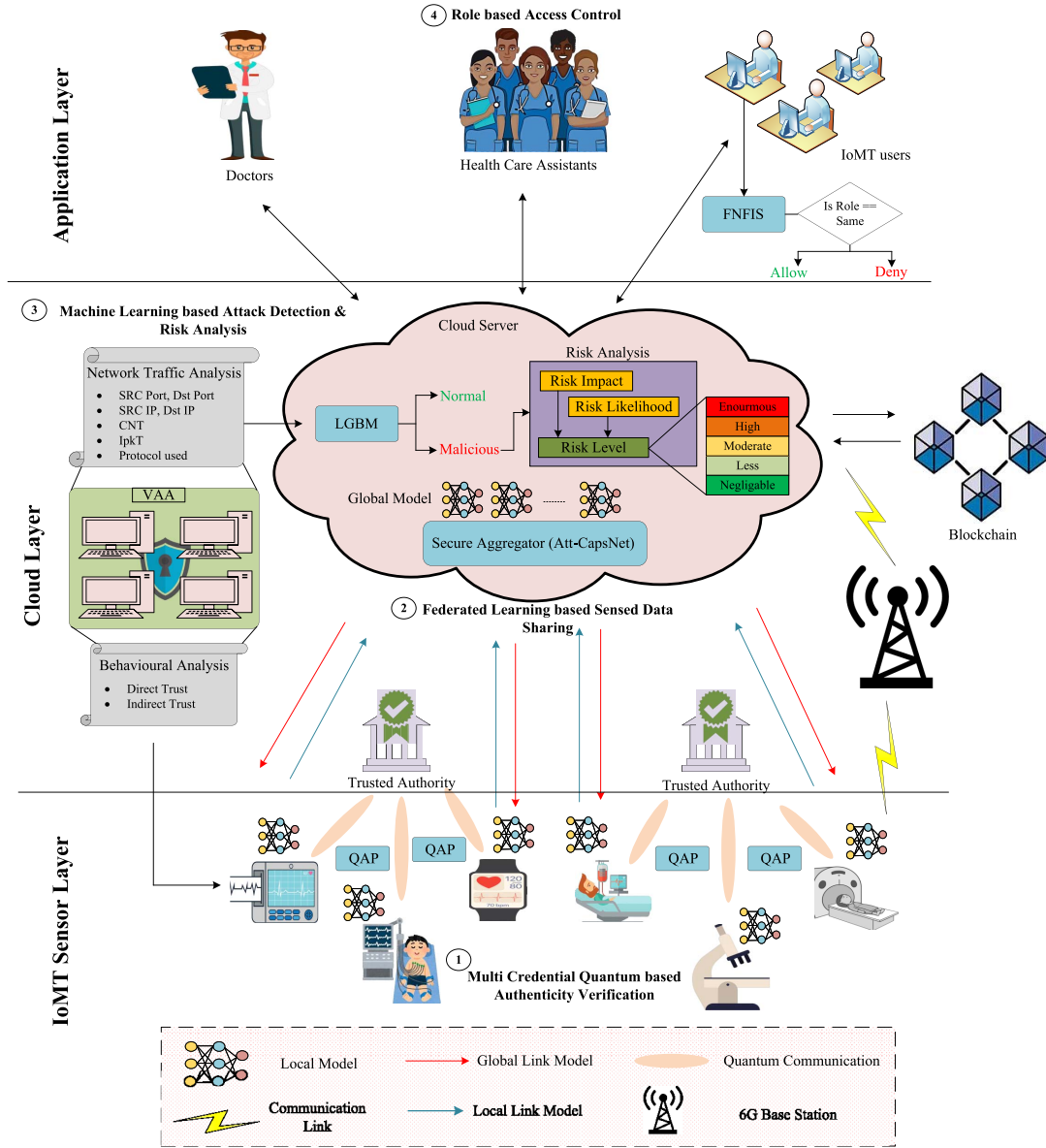


Fig. 1. Overall architecture of the proposed Fed-IoMT block.

blockchain as a conventional participant, TA, and clients (i.e., IoMT devices, doctors, and healthcare assistants). The TA and clients possess quantum capabilities enabling trustworthy authentication.

The blockchain is considered an honest entity, while the TA and clients are classified as partially honest. The primary advantage of QAP is its zero-loss, noise-free authentication process, ensuring high security. The proposed QAP consists of two phases, outlined below:

Phase 1: Enrollment

Step 1: In the initial phase, the TA, clients Cl , including IoMT devices $D_n = \{D_1, D_2, \dots, D_n\}$, doctors $doc_n = \{doc_1, doc_2, \dots, doc_n\}$, and healthcare assistants $ha_n = \{ha_1, ha_2, \dots, ha_n\}$, are registered with their secret credentials ($\mathbb{Z}$) onto the blockchain B . The representation of $\mathbb{Z}$ within the blockchain is formulated as follows:

$$B \leftarrow \begin{cases} \mathbb{Z}_{TA0} = \{\mathbb{Z}_{TA01}, \mathbb{Z}_{TA02}, \dots, \mathbb{Z}_{TA0R}\} \\ \mathbb{Z}_{D1} = \{\mathbb{Z}_{D11}, \mathbb{Z}_{D12}, \dots, \mathbb{Z}_{D1N}\} \\ \mathbb{Z}_{doc2} = \{\mathbb{Z}_{doc21}, \mathbb{Z}_{doc22}, \dots, \mathbb{Z}_{doc2N}\} \\ \mathbb{Z}_{ha3} = \{\mathbb{Z}_{ha31}, \mathbb{Z}_{ha32}, \dots, \mathbb{Z}_{ha3N}\} \end{cases} \quad (1)$$

where, $\mathbb{Z}_{D1j}$, $\mathbb{Z}_{doc2j}$, and $\mathbb{Z}_{ha3j} \in \{0, 1\}$ for $(j = 1, 2, \dots, N)$, and $\mathbb{Z}_{TA0j} \in \{0, 1\}$ for $(j = 1, 2, \dots, R)$. The secret credential $\mathbb{Z}$ consists of its ID, password, biometrics, and assigned roles.

Step 2: Upon successful registration, an authentication token is issued to the TA and clients. The representation of Step 2 is as follows:

$$B \rightarrow A_{Tok}[D, doc, ha] \quad (2)$$

where A_{Tok} denotes the authenticity token, serving as a credential for clients to initiate the authentication process.

Phase 2: Authenticity Verification

Step 3: The clients initiate authentication by submitting A_{Tok} to the TA. Upon receiving the clients' A_{Tok} tokens, the TA generates random quantum sequences of $R(N+1)$ GHZ quantum states, formulated as follows:

$$\begin{cases} |\mathbb{G}_1\rangle = \frac{1}{\sqrt{2}}(|\mathbb{S}_{TA_{01}}\mathbb{S}_{D_{11}}\dots\mathbb{S}_{D_{n1}}\rangle \\ + |(\mathbb{S}_{TA_{01}}\oplus 1)(\mathbb{S}_{D_{11}}\oplus 1)\dots(\mathbb{S}_{D_{n1}}\oplus 1)\rangle) \\ |\mathbb{G}_2\rangle = \frac{1}{\sqrt{2}}(|\mathbb{S}_{TA_{02}}\mathbb{S}_{doc_{12}}\dots\mathbb{S}_{doc_{n2}}\rangle \\ + |(\mathbb{S}_{TA_{02}}\oplus 1)(\mathbb{S}_{doc_{12}}\oplus 1)\dots(\mathbb{S}_{doc_{n2}}\oplus 1)\rangle) \\ |\mathbb{G}_3\rangle = \frac{1}{\sqrt{2}}(|\mathbb{S}_{TA_{03}}\mathbb{S}_{ha_{21}}\dots\mathbb{S}_{D_{n3}}\rangle \\ + |(\mathbb{S}_{TA_{03}}\oplus 1)(\mathbb{S}_{ha_{21}}\oplus 1)\dots(\mathbb{S}_{ha_{n3}}\oplus 1)\rangle) \end{cases} \quad (3)$$

where $TA_{0n}, D_{1n}, doc_{2n}, ha_{3n}$ represent the $(N+1)$ quantum particles of the n th GHZ state.

Step 4: After generating the random quantum sequences, the TA splits the GHZ sequences into ordered sequences: $\mathbb{S}_{TA_0}, \mathbb{S}_{D_1}, \mathbb{S}_{doc_2}, \mathbb{S}_{ha_3}$. The TA then generates quantum decoy photons $\{|0\rangle, |1\rangle, |+\rangle, |-\rangle\}$ and inserts them into the ordered sequences. The TA retains $\mathbb{S}_{TA_0}$ and provides $\mathbb{S}_{D_1}, \mathbb{S}_{doc_2}, \mathbb{S}_{ha_3}$ to the clients.

Step 5: Upon receiving the ordered sequences $Seq = \{\mathbb{S}_{D_1}, \mathbb{S}_{doc_2}, \mathbb{S}_{ha_3}\}$, the clients perform quantum operations, specifically GNZ and XOR operations, on the quantum decoy photons. This process is formulated as follows:

$$Seq' = GNZ[\mathbb{S}_{D_1}] \oplus GNZ[\mathbb{S}_{doc_2}] \oplus GNZ[\mathbb{S}_{ha_3}] \quad (4)$$

$$Cl[Seq'] \rightarrow TA \quad (5)$$

Step 6: Upon receiving Seq' from the clients, the TA performs the same operations as in Step 5 to retrieve the original sequences. The correctness of the sequences is verified by computing the error between the original sequences held by the TA and those received from the clients. The error computation is formulated as follows:

$$Err = Seq - Seq' \quad (6)$$

In the above equation, a threshold value (Th) is set by the blockchain to verify client authenticity. If the computed error is greater than or equal to the threshold, the client is considered malicious; otherwise, the client is considered legitimate. The formulation for authenticity verification is expressed as:

$$TA = \begin{cases} \text{Discarded,} & \text{if } Err \geq Th \\ \text{Authenticated,} & \text{otherwise} \end{cases} \quad (7)$$

Similarly, clients compute the error for the TA and publish the results to the blockchain.

Step 7: The blockchain determines the final decision regarding whether the communication between the clients and the TA is legitimate or malicious. The results are securely stored as blocks within the blockchain.

B. Federated Learning-Based Sensed Data Sharing

Only authenticated IoMT devices are allowed to participate in the IoMT environment, specifically the authenticated patients' IoMT devices for sensing. To enhance user privacy, the proposed framework employs federated learning and homomorphic encryption. For each round r , IoMT devices

acquire a global model from the cloud server, train and encrypt their local models, compute decryption dividends, and perform secure aggregation. This process is repeated until convergence is achieved. A detailed explanation of the proposed FL-based data-sharing process is provided below.

First, the server initializes the public key parameters for each IoMT device based on its authenticity $d(\forall, \delta, \Upsilon, \eta, \Lambda)$, where Λ is a random vector, $\Lambda \leftarrow U(T_V^D)$, Υ and η represent the error distribution over T , δ denotes the key distribution parameter, and $\forall$ is the modulus of ciphertext. Based on the public key parameters, each IoMT device D_j generates a secret key $sec_{kj} \leftarrow \delta$, from which the public key is computed as $Pk_j = -sec_{kj} \cdot \Lambda + e_j(\text{mod } \forall)$. Using this, all authenticated IoMT devices compute the collaborative public key as follows:

$$\tilde{P}k = \sum_{j=1}^n Pk_j = \sum_{j=1}^n (-sec_{kj}) \cdot \Lambda + \sum_{j=1}^n e_j (\text{mod } \forall) \quad (8)$$

Once the collaborative public key is computed, for each round r , every IoMT device D_j downloads the current weights of the global model we_t from the cloud server. Using these weights, each device generates a local model by employing the Att-CapsNet algorithm. The integration of a squeeze-and-excitation block into a conventional CapsNet serves as an attention mechanism that enhances the learning of global feature representations. This incorporation addresses dimensionality challenges, resulting in improved channel weight estimation.

Local Model Generation: Initially, the sensed data from the IoMT devices are fed into the convolutional layers of the Att-CapsNet. In these layers, local features are extracted using convolutional kernels of size 3×3 with a stride of one. The extracted feature maps are then passed to the squeeze-excitation layer. The squeeze-excitation network consists of two key operations: squeeze and excitation, which extract global features.

The squeeze operation extracts features to generate the feature map, while the excitation operation determines the channel weights based on the computed feature maps. The formulation of the squeeze-excitation operation is as follows:

$$Q_c = Seq(\mathbb{S}_p) = \frac{1}{H \times W} \sum_{j=1}^H \sum_{i=1}^W \mathbb{S}_p(j, i) \quad (9)$$

$$Sc = \text{Exi}(Q, W) = \mathbb{Z}(A(Q, W)) = \mathbb{Z}(W_2 \varphi(W_1 Q)) \quad (10)$$

$$\tilde{y}_p = \text{scale}(\mathbb{S}_p, Sc_p) = Sc_p \cdot \mathbb{S}_p \quad (11)$$

From equations (9)–(11), $\mathbb{S}_p$ represents the p th feature map, W_2 and W_1 denote the dimensionality weights, Sc represents the channel weight, and Q_c denotes the statistical representation of the channel. The generated feature maps from the squeeze-excitation layer are then passed to the primary capsule layer, where they are transformed into primary capsules. The capsules store a large amount of information in vector format. These vector-formatted capsules are subsequently provided as inputs to the digit capsule layer, which produces the final output.

The digit capsule layer incorporates a squashing function to facilitate vector-directed routing. The routing process between the j th and i th capsules is formulated as follows:

$$\hat{\mathbf{s}}_{ij} = W_{ji} \mathbf{s}_j \quad (12)$$

$$S_{ci} = \sum_j \partial_{ji} \hat{\mathbf{s}}_{ij} \quad (13)$$

$$O_i = \frac{\|S_{ci}\|^2}{1 + \|S_{ci}\|^2} \cdot \frac{S_{ci}}{\|S_{ci}\|} \quad (14)$$

From Equations (12)–(14), W_{ji} represents the matrix weight emphasizing the relationship between the j th and i th capsules, ∂_{ji} denotes the connection coefficient, and O_i refers to the output squashing function that provides the sensed results of individual patients. Once the local model is derived after several epochs, IoMT devices provide the weights of each individual local model as $we_t^j, j \in 1, 2, \dots, n$.

Encrypting the Local Model Weights: The plaintext input for we_t^j is represented as $f_j \in T$. Let $\Lambda = \Lambda[0]$ and $\tilde{P}k = \tilde{P}k[0]$. The samples from the public key parameters are given as $\vartheta^{D_i} \leftarrow \delta$, and $e_0^{D_i}, e_1^{D_i} \leftarrow \Upsilon$. The ciphertext for the local model is computed as follows:

$$\text{CpT}^{D_i} = (\vartheta^{D_i} \cdot \tilde{P}k + f_j + e_0^{D_i}, \vartheta^{D_i} \cdot \Lambda + e_1^{D_i}) \pmod{\forall} \quad (15)$$

From the above equation, each IoMT device shares the encrypted local model $\text{CpT}^{D_i} = (\text{CpT}_0^{D_i}, \text{CpT}_1^{D_i})$ with the cloud server.

Perform Homomorphic Operation: Homomorphic verification protects cloud-stored data from cyberattacks and enables data auditing without downloading the entire encrypted dataset. To improve verification, homomorphic verification incorporates non-malleability and Context Binding. Non-malleability protects legitimate ciphertext from generation or modification by attackers, while Context Binding prevents fake data auditing at the cloud server. Once the server receives the encrypted local model weights from all IoMT devices, it performs homomorphic encryption operations by summing all the weights. The formulation of the homomorphic operation on the local model weights is as follows:

$$\begin{aligned} \text{CpT}_{\text{sum}} &= \sum_{j=1}^N \text{CpT}^{D_i} = \left(\sum_{j=1}^N \text{CpT}_0^{D_i}, \sum_{j=1}^N \text{CpT}_1^{D_i} \right) \\ &\triangleq (\text{CpT}_{\text{sum}_0}, \text{CpT}_{\text{sum}_1}) \pmod{\forall} \end{aligned} \quad (16)$$

where CpT_{sum} is the sum of homomorphic operations of ciphertexts.

Decryption Dividends Computation: The homomorphically encrypted ciphertext sum is distributed to all IoMT devices to compute the decryption dividends, which are then sent back to the cloud server. The formulation for decryption dividends is as follows:

$$\begin{aligned} D_i &= \text{sec}_{ki} \cdot \text{CpT}_{\text{sum}_1} + e_j^* = \\ &\text{sec}_{kj} \cdot \sum_{j=1}^N (\vartheta_i \cdot \Lambda + e_1^{D_i}) + e_1^* \pmod{\forall} \end{aligned} \quad (17)$$

Secure Global Aggregation: Upon receiving all decryption dividends from IoMT devices, the cloud server performs

Algorithm 1 FL-Based Data Sharing

Input: Authenticated IoMT devices D_i

Output: Aggregated Sensed Model with new model weight we_{t+1}

```

1: Initialize collaborative  $\tilde{P}k$  (8)
2: for every round  $r \rightarrow D_j$  do
3:   Every  $D_j[\text{get}(we_t)]$  from the cloud server
   // Local Model Generation //
4:   Sensed results  $\rightarrow \text{AttCapsNet}[\text{Conv}(3 \times 3)]$ 
5:    $\text{AttCapsNet}[\text{Conv}(3 \times 3)]$  extracts local features  $L_{\text{fea}}$ 
6:    $L_{\text{fea}} \rightarrow \text{Seq}(\cdot)\text{Exi}(\cdot)$  for  $S_{cp} \cdot \mathbf{s}_p$  (9)–(11)
7:    $\mathbf{s}_p \rightarrow$  primary capsules for vector format conversion
8:   OP (primary capsules)  $\rightarrow$  Digit capsules (12)–(14)
9:   Assign weight for every IoMT local model  $we_t^j$ 
   // Encrypt Local Model Weights //
10:  Represent  $we_t^j$  to plaintext form  $f_j \in T$ 
11:  With public-key parameters compute  $\text{CpT}^{D_i}$  (15) to cloud server
   // Perform Homomorphic Operation //
12:  Apply HE  $\text{CpT}_{\text{sum}}$  on  $\text{CpT}^{D_i}$  (16)
13:   $\text{CpT}_{\text{sum}} \rightarrow D_i$  for computing decryption dividends (17)
   // Secure Global Aggregation //
14:  Decryption dividends  $\rightarrow$  Cloud server
15:  Perform secure local model aggregation (18)
16:  Assign a new global model weight  $we_{t+1}$  (19)
17:  Run until max  $r$  met
18: end for
```

secure aggregation to obtain the plaintext. The formulation for secure aggregation is as follows:

$$\sum_{i=1}^N f_j \approx \text{CpT}_{\text{sum}_0} + \sum_{i=1}^N D_i \pmod{\forall} \quad (18)$$

From the above equation, the cloud server decrypts $\sum_{i=1}^N f_j$ to compute the average local weights and obtain the new global model weight we_{t+1} for further rounds.

$$we_{t+1} = \frac{1}{N} \sum_{j=1}^N we_t^j \quad (19)$$

The communication rounds continue until the desired convergence is met. The pseudocode for Federated Learning-based data sharing is given in Algorithm 1.

C. Machine Learning-Entrenched Attack Detection and Risk Analysis

The Vulnerability Analysis Agent (VAA) in the cloud server monitors the communication between IoMT devices and the cloud server. Continuous monitoring is enabled by observing the behaviors of IoMT devices and network traffic to the cloud server. Specifically, the behavior of IoMT devices is captured by computing both direct and indirect trust measures. Network flows are analyzed by examining flow and packet features.

1) *Behavioral Analysis:* For each IoMT device D_j in each communication round r , the VAA computes direct and indirect trust (dir_{tru}) and ($indir_{tru}$), respectively. dir_{tru} is computed by analyzing the past behaviors of a particular IoMT device. The formulation of dir_{tru} is as follows:

$$dir_{tru}[D_j] = Z \cdot \mathcal{Z}^y \quad (20)$$

where Z represents the past behavior of device D_j , and $\mathcal{Z}^y$ denotes the current behavior monitored by the VAA. Indirect trust, $indir_{tru}$, is computed as follows:

Algorithm 2 Packet Flow Analysis

```

1: BEGIN
  // Extracting source and destination port details //
2: SRC_Port ← Packet.Header.Source_Port
3: DEST_Port ← Packet.Header.Destination_Port
  // Identifying transport layer protocol //
4: Protocol ← Packet.Header.Transport_Protocol
  // Extracting IP addresses //
5: SRC_IP ← Packet.Header.Source_IP
6: DEST_IP ← Packet.Header.Destination_IP
  // Monitoring packet exchange statistics //
7: CNT ← Compute_Byte_Packet_Count(Packet)
  // Calculating inter-packet arrival time //
8: IPkt ← Compute_Inter_Packet_Time(Cloud_Server, Device)
9: END

```

$$indir_{tru}[D_j] = \frac{1}{nr} \sum_{i=1}^n Fe[D_i] \rightarrow D_j \quad (21)$$

where nr denotes the number of feedbacks obtained about the behavior of device D_j , and $Fe[D_i]$ represents the feedback from device D_i regarding device D_j . Finally, the overall trust value is computed by summing dir_{tru} and $indir_{tru}$ as follows:

$$Tr = dir_{tru}[D_j] + indir_{tru}[D_j] \quad (22)$$

2) *Network Traffic Analysis*: Simultaneously, the flows from IoMT devices D_j , as well as external malicious devices, are analyzed by considering several features. The features extracted in the proposed work are listed below and also described in Algorithm 2:

- SRC_Port: The source port extracted from the packet header; used by the sending peer.
- DEST_Port: The destination port from the packet header; used by the receiving peer.
- Protocol Used: Specifies the transport layer protocol in use (e.g., UDP, TCP).
- DES_IP: The destination IP address of the receiving peer.
- SRC_IP: The source IP address of the transmitting peer.
- CNT: Represents the number of bytes or packets exchanged during a defined time interval.
- IPkt: The inter-packet arrival time, indicating the time between packets during flow exchange between the cloud server and devices.

By obtaining these features, the analysis is performed using several feature aggregation techniques, such as packet and byte counts over a particular second, standard deviation computation for flow, bytes, and packets, and standard deviation measures among byte-to-duration counts. The analyzed network traffic for all devices is represented as $Ntr_m = \{Ntr_1, Ntr_2, \dots, Ntr_m\}$.

3) *LGBM-Based Risk Analysis and Prevention*: The LGBM is a powerful machine-learning classifier for complex decision-making situations. The proposed method trains the classifier using trust and traffic analysis results from the devices. The training set is represented as $\{(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)\}$, where the given data samples are represented as a , and the labels for classification are represented as b . The goal of the Light Gradient Boosting Machine is to minimize the loss function during classification.

Algorithm 3 LGBM-Based Device Classification

```

Input:  $Tr(D_j), Ntr_m$ 
Output: Classification Result (Normal or Malicious)
1: for all  $D_j$  do
2:   Set training samples:  $\{(a_1, b_1), (a_2, b_2), \dots, (a_n, b_n)\}$ 
3:   Perform Loss Minimization:  $\widehat{F(a)}$  (23)
4:   Determine Iterative Function (24)
5:   Provide Final Classification:  $F_U(a)$  (25)
6:   Provide results:  $\omega_u(a)$  (26)
7: end for
8: for all Malicious  $D_j$  do
9:   Analyze Risk( $D_j$ ) (27)
10:  Formulate Risk Impact:  $Imp(D_j)$  (28)
11:  Formulate Risk Likelihood:  $lik(D_j)$  (29)
12:  Calculate:  $Overall Risk \leftarrow Imp(D_j) \times lik(D_j)$ 
13:  if Risk score > 90 then
14:    Critical Impact
15:  else if Risk score  $\geq 70$  and < 90 then
16:    High Impact
17:  else if Risk score  $\geq 60$  and < 70 then
18:    Moderate Impact
19:  else if Risk score  $\geq 20$  and < 60 then
20:    Low Impact
21:  else
22:    Negligible Impact
23:  end if
24: end for

```

The loss can be minimized using $F(a)$, which is formulated as follows:

$$\widehat{F(a)} = \arg \min_F \varepsilon_{a,b}[\mathbb{L}(b, F(a))] \quad (23)$$

From the above equation, the iterative function of the LGBM can be used to minimize the loss function as follows:

$$F_u(a) = F_{u-1}(a) + \alpha_u \varpi_u(a) \quad (24)$$

where u defines the number of iterations, $\varpi_u(a)$ denotes the decision tree base, and $\alpha_u = \arg \min_{\alpha} \sum_{i=1}^n \mathbb{L}(b_i, F_{u-1}(a_i) + \alpha \varpi_u(a_i))$. From this equation, the final classification model for the LGBM is obtained by combining the weights as follows:

$$F_U(a) = \sum_{u=1}^U \alpha_u \varpi_u(a) \quad (25)$$

The maximized iteration number is denoted by U , and the final classification from the base decision tree $\varpi_u(a)$ can be formulated as follows:

$$\varpi_u(a) = \begin{cases} \text{Normal } D_j & \text{if condition} \\ \text{Malicious } D_j & \text{otherwise} \end{cases} \quad (26)$$

From the classifier results, normal D_j values are accepted, while malicious D_j values are subjected to risk analysis. The pseudocode defining the operation of the LGBM device classification and risk analysis is shown in Algorithm 3.

To analyze the risk of any IoMT device D_j , the formulation is defined as follows:

$$Risk(D_j) = Imp(D_j) \times lik(D_j) \quad (27)$$

where $Imp(D_j)$ defines the impact of risk and $lik(D_j)$ defines the likelihood of the risk. The impact of risk is determined based on metrics such as device security, considering whether an attack compromises confidentiality, integrity, or availability, the type of protocol used, the network type, and the number of heterogeneous IoMT devices involved.

TABLE I
SECURITY RISK ANALYSIS FOR IOMT DEVICES

IoMT Type	Attack	sec(D_j)	CIA(D_j)	Pro(D_j)	NT(D_j)	Het(D_j)	Imp(D_j)	RF(D_j)	His(D_j)	Sec(D_j)	Lay(D_j)	lik(D_j)	Level of Risk
Smart Wearables	Spoofing	8	10	8	5	10	8.2	7	9.2	10	6	8.05	66.01 (Moderate)
Sugar Level Monitor	DOS	9	9	6.5	4	10	7.7	10	10	8	9	9.25	71.23 (High)
Smart Bed	MITM	5	4	3	6	2	4	6	6.6	7	5	6.15	24.6 (Less)

$$\text{Imp}(D_j) = [\text{sec}(D_j) + \text{CIA}(D_j) + \text{Pro}(D_j) + \text{NT}(D_j) + \text{Het}(D_j)]/5 \quad (28)$$

where $\text{sec}(D_j)$, $\text{CIA}(D_j)$, $\text{Pro}(D_j)$, $\text{NT}(D_j)$, and $\text{Het}(D_j)$ denote the security, CIA, protocol used, network type, and number of heterogeneous devices, respectively. The likelihood of risk can be formulated as follows:

$$\text{lik}(D_j) = [\text{RF}(D_j) + \text{His}(D_j) + \text{Sec}(D_j) + \text{Lay}(D_j)]/4 \quad (29)$$

where $\text{RF}(D_j)$ represents the risk factor, $\text{His}(D_j)$ denotes the history of attacks encountered by the IoMT device, $\text{Sec}(D_j)$ refers to the sector type, and $\text{Lay}(D_j)$ represents the proposed IoMT layer, which is more prone to cyber-attacks. Accordingly, the risk level is classified into five categories: critical impact, high impact, moderate impact, low impact, and negligible impact. Table I presents the risk impact analysis of the IoMT devices in the proposed environment, with a detailed description available in Algorithm 3.

D. Adaptive Fuzzy Role-Based Access Control

Access control is provided for IoMT users, doctors, and healthcare assistants in cloud applications using the Flexible Neuro-Fuzzy Inference System (FNFIS), based on the roles assigned during the authentication stage. For instance, doctors interpret and provide results for IoMT users, IoMT users view the final results provided by doctors, and healthcare assistants manage IoMT users' requests without compromising their privacy. Entities that comply with role-based policies are granted access; otherwise, access is denied.

The roles of the entities are input to the FNFIS, which consists of five layers: fuzzification, inference, normalization, defuzzification, and output. The entity roles are represented as $Ro = \{Ro(D_n), Ro(doc_n), Ro(ha_n)\}$. The output of the fuzzification layer is expressed as follows:

$$Op_I^i = \begin{cases} \sqsupset_{\mathbb{A}_i}(x), & \forall i = 1, 2 \\ \sqsupset_{\mathbb{B}_{i-2}}(x), & \forall i = 3, 4 \\ \sqsupset_{\mathbb{C}_{i-4}}(x), & \forall i = 5, 6 \end{cases} \quad (30)$$

From the above equation, the fuzzy membership function is denoted by $\sqsupset$, and the linguistic variables are denoted by $\mathbb{A}_i$, $\mathbb{B}_i$, and $\mathbb{C}_i$. The proposed FNFIS uses a Gaussian function, formulated as follows:

$$f(x) = \mathfrak{A} \cdot \exp\left\{-\frac{(x - \mathfrak{B})^2}{2\mathfrak{C}^2}\right\} \quad (31)$$

where $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}$ denote the membership parameters of the FNFIS. These parameters are provided to the inference layer, which produces the output as follows:

$$Op_{II}^i = We_i = \sqsupset_{\mathbb{A}_i}(x) \cdot \sqsupset_{\mathbb{B}_i}(x) \cdot \sqsupset_{\mathbb{C}_i}(x), \quad \forall i = 1, 2, 3 \quad (32)$$

Algorithm 4 Role-Based Access Control

Input: Client Roles $\{Ro(D_n), Ro(doc_n), Ro(ha_n)\}$

Output: FNFIS Access Rules 1 or 0

```

1: Initialize the Roles and Authenticity of  $Cl$ 
2: Pass  $Ro(Cl) \rightarrow$  fuzzification layer (30)
3: Set the FNFIS Gaussian Function  $f(x)$  (31)
4:  $Op_I^i \rightarrow$  Inference Layer to determine weights (32)
5: Normalize  $We_i$  in the normalization layer (33)
6: Defuzzify the normalized weights in the de-fuzzification layer (34)
7: Obtain overall output  $Op_V^i$  (35)
8: Generate Fuzzy Rules from  $Op_V^i$ 
9: if  $Ro(Cl) = Ac(res)$  then
10:   Allowed = 1
11: else
12:   Denied = 0
13: end if

```

From the above equation, the weight of a neuron is denoted by We_i . The weights from the inference layer are normalized in the normalization layer, which is formulated as follows:

$$Op_{III}^i = \overline{We}_i = \frac{We_i}{\sum_i We_i}, \quad \forall i = 1, 2, 3 \quad (33)$$

From the above equation, the normalized weight is denoted by $\overline{We}_i$. The normalized weights are defuzzified in the defuzzification layer, formulated as follows:

$$Op_{IV}^i = \overline{We}_i f_i = \overline{We}_i \cdot (\mathfrak{P}_i x + \mathfrak{Q}_i y + \mathfrak{S}_i z) \quad (34)$$

where $\mathfrak{P}$, $\mathfrak{Q}$, and $\mathfrak{S}$ denote the subsequent fuzzy parameters that provide the defuzzified output. The overall output from the output layer is formulated as follows:

$$Op_V^i = f(x, y, z) = \sum_i \overline{We}_i f_i = \frac{\sum_i \overline{We}_i f_i}{\sum_i \overline{We}_i} \quad (35)$$

The final output from the output layer provides access rules for clients based on their designated roles in Boolean format. The formulation of the access rules is as follows:

$$\text{RULE(FNFIS)} = \begin{cases} 1, & \text{if } Ro(Cl) = Ac(res) \\ 0, & \text{if } Ro(Cl) \neq Ac(res) \end{cases} \quad (36)$$

where 1 denotes access granted and 0 denotes access denied. Specifically, if the client roles and their access requests $Ac(res)$ match, access is granted (1); otherwise, it is denied (0). The pseudocode for the FNFIS-assisted role-based access control is provided in Algorithm 4.

V. EXPERIMENTAL RESULTS

This section presents the experimental and implementation analyses of the proposed Fed-IoMT-Block framework, offering both quantitative and qualitative insights. It consists of three supplementary sections, each described in detail below:

TABLE II
SIMULATION SETTINGS

Simulation Parameters	Description
# of IoMT devices	100
# of IoMT users	20
# of Cloud Server	1
# of Blockchain Node	1
# of Agents	2
Nature of IoMT devices	Heterogeneous
Placement Status	Random
Block Size	4 bytes
Transaction Size	8 bytes
Simulation Area	1500m × 1500m
Simulation Time	150s
Security Protocol	Homomorphic Encryption
Network Traffic	Transmission Control Protocol

A. Simulation Setup

The simulation setup for the proposed Fed-IoMT-Block framework is detailed in this section. The framework was implemented using Python 3.8.2 on Windows 11, with 8GB of RAM, 500GB HDD storage, and an AMD Ryzen 5 5600H microprocessor with Radeon Graphics at 3.30 GHz. Table II lists the simulation parameters of the proposed method.

The heterogeneous nature of IoMT devices includes on-body, in-body, and off-body sensing devices, which are used to monitor respiratory rate, heart rate, blood pressure, temperature, internal organs such as the kidneys, and the physical postural movements of the patient, respectively. The sensory data generated by these devices require different transmission speeds and bandwidths to be periodically sent to medical doctors.

B. Comparative Analysis

This section focuses on the comparative analysis of the proposed Fed-IoMT-Block with existing frameworks, including the Fortified Chain [31], Fed-IoMT [38], and ASCP-IoMT [46]. The performance metrics used for quantitative analysis include Malicious Traffic (bps), Detection Accuracy (%), Attack Detection Accuracy (%), and Throughput (Mbps). A detailed explanation of the comparative metrics is provided below.

1) *Malicious Traffic Analysis*: Malicious traffic is defined as traffic initiated or imposed by malicious users/entities in the IoMT environment, attempting to take over legitimate resources in an untrusted environment. Figure 2 compares malicious traffic in the proposed Fed-IoMT-Block framework with existing methods based on the number of IoMT devices. It is observed that the malicious traffic generated by attackers is lower in the proposed framework than in existing solutions. The primary reason for this reduction is that the proposed framework filters such traffic early by performing authentication using the QAP protocol. The QAP utilizes multiple authentication metrics, including ID, password, biometrics, and user roles, to establish a robust authentication policy. Additionally, quantum communication helps prevent communication-related attacks commonly found in conventional cryptography-based authentication methods.

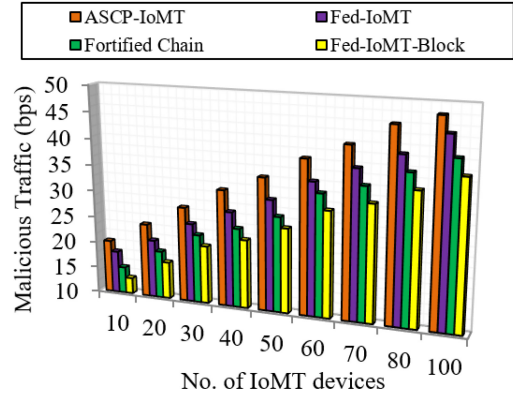


Fig. 2. Number of IoMT devices vs malicious traffic.

TABLE III
MALICIOUS TRAFFIC RATE (BPS) FOR IOMT DEVICES

IoMT Devices	Fed-IoMT-Block	Fortified Chain	Fed-IoMT	ASCP-IoMT	Difference (bps)
Min 10	13	15	18	20	2–7
Max 100	38	41	45	48	3–10
Average	26.12	28.56	31.34	34.89	2.44–8.77

In contrast, existing methods such as the Fortified Chain, Fed-IoMT, and ASCP-IoMT either rely on conventional cryptography-based authentication or lack efficient authentication mechanisms, resulting in higher malicious traffic rates. The quantitative analysis in Figure 2 shows that as the number of IoMT devices increases to a maximum of 100, the proposed Fed-IoMT-Block achieves a lower malicious traffic rate of 38 bps. In comparison, the existing solutions, including Fortified Chain, Fed-IoMT, and ASCP-IoMT, experience higher rates of 41 bps, 45 bps, and 48 bps, respectively. Similarly, when the number of IoMT devices is at a minimum of 10, the proposed Fed-IoMT-Block records 13 bps, while Fortified Chain, Fed-IoMT, and ASCP-IoMT record 15 bps, 18 bps, and 20 bps, respectively. Table III presents the detailed comparison.

2) *Detection Accuracy Analysis*: Detection accuracy is the proportion of true positives to the sum of true positives and true negatives. In our work, we define detection accuracy as the correct identification of disease by heterogeneous IoMT devices, as follows.

$$\text{Det}^{\text{Acc}} = \frac{TP}{TP + TN} \times 100 \quad (37)$$

Figure 3 shows detection accuracy in relation to disease severity. The graphical analysis indicates that detection accuracy increases as disease severity increases.

The proposed Fed-IoMT-Block achieves improved detection accuracy by utilizing an advanced AI model for disease detection called Att-CapsNet. This model outperforms the conventional CNN algorithm and ensures higher detection accuracy. In contrast, existing works such as Fortified Chain, Fed-IoMT, and ASCP-IoMT achieve lower detection accuracy due to their reliance on conventional machine learning classifiers. The numerical results show that the detection accuracy of the proposed Fed-IoMT-Block increases to 98%

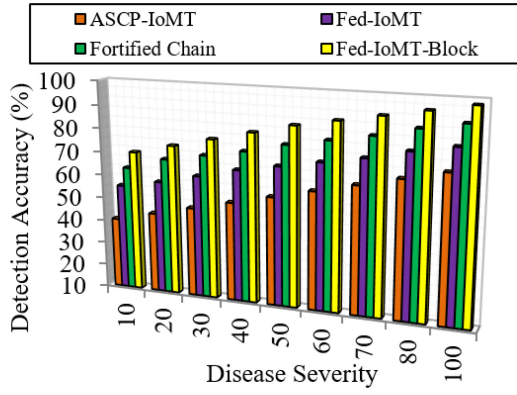


Fig. 3. Disease severity vs detection accuracy.

TABLE IV
DETECTION ACCURACY (%) FOR IoMT DEVICES

IoMT Devices	Fed-IoMT-Block	Fortified Chain	Fed-IoMT	ASCP-IoMT	Difference (%)
Min 10	70	63	55	40	7–30
Max 100	98	91	82	72	7–16
Average	84.89	77.56	68.67	56	28.89–7.33

when disease severity reaches its maximum of 100. However, existing solutions achieve detection accuracy rates of 91%, 82%, and 72%, respectively. Similarly, when disease severity is a minimum of 10, the detection accuracies of the proposed and existing methods are 70%, 63%, 55%, and 40%, respectively. Table IV provides a better representation of the detection accuracy.

3) *Attack Detection Accuracy Analysis*: Attack detection accuracy is defined as the proportion of legitimate sample counts in the environment to the number of malicious samples, as described below:

$$\text{AttDec} = \frac{\text{No. of Legitimate Samples}}{\text{No. of Malicious Samples}} \times 100 \quad (38)$$

Figure 4 shows that the proposed Fed-IoMT-Block achieves a higher attack detection accuracy rate than existing methods. The primary reason for this improved detection rate is the incorporation of a VAA, which continuously monitors external attacks. In addition, the proposed framework employs a blockchain-based role-based access control method, further strengthening the security of the Fed-IoMT-Block model.

Quantitative analysis indicates that when the number of malicious IoMT devices increases to a maximum of 50, the proposed Fed-IoMT-Block detects attacks at a higher rate of 99%, whereas existing solutions achieve lower detection rates of 86%, 83%, and 79%, respectively. Similarly, when the number of malicious IoMT devices is as low as 10, the proposed framework achieves a higher attack detection accuracy of 65%, while existing solutions achieve detection rates of 60%, 53%, and 40%, respectively. Table V shows the attack detection accuracy of the proposed Fed-IoMT-Block framework compared to existing state-of-the-art schemes, achieving a higher attack detection accuracy.

4) *Throughput Analysis*: Throughput is defined as the number of packets successfully transmitted from the source to the

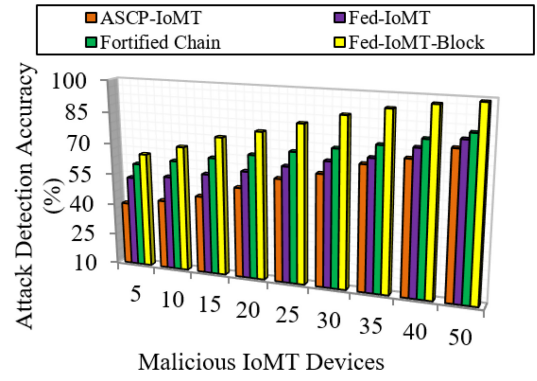


Fig. 4. Malicious IoMT devices vs attack detection accuracy.

TABLE V
ATTACK DETECTION ACCURACY (%) FOR IoMT DEVICES

IoMT Devices	Fed-IoMT-Block	Fortified Chain	Fed-IoMT	ASCP-IoMT	Difference (%)
Min 10	65	60	53	40	5–25
Max 50	99	86	83	79	13–20
Average	84	72.34	66	58.45	11.6–25.55

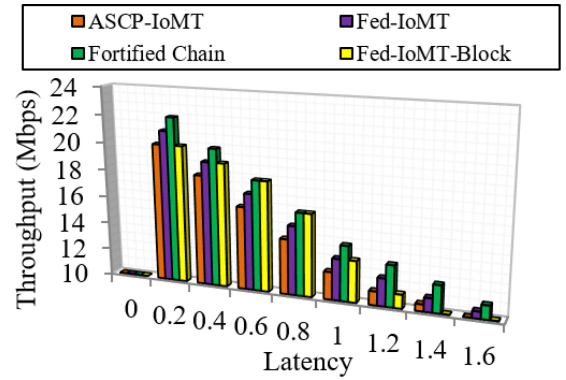


Fig. 5. Latency vs throughput.

destination in the IoMT environment. Figure 5 shows that as latency increases, throughput decreases because it is inversely proportional to latency. The proposed framework achieves better throughput, even as latency increases. The primary reason for this higher throughput rate is that the proposed framework incorporates federated learning technology into the IoMT environment, ensuring data privacy that helps resist communication-related attacks and protect model privacy. In contrast, existing solutions such as Fortified Chain, Fed-IoMT, and ASCP-IoMT have limited data privacy measures, resulting in lower throughput.

Based on the numerical analysis, the proposed framework achieves a higher throughput rate of 12 Mbps when latency increases to a maximum of 1.6 seconds. For the same latency, the throughput achieved by Fortified Chain, Fed-IoMT, and ASCP-IoMT is 11 Mbps, 10.5 Mbps, and 10 Mbps, respectively. Similarly, when latency is a minimum of 0.2 seconds, the proposed framework achieves a throughput of 23 Mbps, while existing solutions achieve 22 Mbps, 21 Mbps, and 20 Mbps, respectively.

TABLE VI
ALGORITHM FUNCTIONS AND COST COMPLEXITY IN THE PROPOSED FED-IOMT-BLOCK FRAMEWORK

Algorithm	Function	Complexity
Quantum Authentication Protocol (QAP)	Secures data transmission by validating device authenticity	$(O(\log N))^2$
Attention Capsule Network (Att-CapsNet)	Generates federated disease models from heterogeneous IoMT data	$O(N + PC + DC)$
Light Gradient Boosting Machine (LGBM)	Classifies network traffic as normal or malicious	$O(0.5 \times \text{fea} \times \text{bin})$
Flexible Neuro-Fuzzy Inference System (FN FIS)	Enables intelligent role-based access control decisions	$O(\text{MF} + \text{Fu} + \text{Nor} + \text{Defu})$

C. Security Analysis

This section presents a security analysis of the proposed countermeasures. Below are the cyber-attacks that were resisted by the proposed method:

- **Man-in-the-Middle Attack (MITM):** The proposed work mitigates this MITM attack by adopting quantum technology-based authentication, named QAP, which overcomes the problems of conventional cryptography algorithms.
- **DDoS Attacks:** The Distributed Denial of Service (DDoS) attack floods legitimate servers to prevent access from legitimate users. DDoS attacks are overcome in the proposed environment by utilizing VAA, which analyzes the risk of external traffic to enable better attack detection accuracy.
- **Insider Attacks:** The proposed architecture consists of three layers including the IoMT Sensors Layer, the Cloud Layer, and the Application Layer. Each layer includes various sensing devices and healthcare personnel. Although these layers are partially secured, they remain vulnerable to compromise. Insider attacks may exploit vulnerabilities through phishing, malware, DoS, and DDoS attacks, leading to the deletion or corruption of sensory data stored on local machines. These incidents are often attributed to human error or hardware/software failures. Any TA, VAA, or SA can be compromised by such attacks and may exhibit Byzantine failures to conceal the loss or corruption of data from medical professionals. Furthermore, insider threats may also involve reassigning stored data to a new user's storage location, effectively removing the original data. To address these challenges, the proposed work adopts role-based access control using FN FIS, which restricts privileged access and effectively mitigates insider threats.
- **Model Poisoning Attacks:** Model poisoning attacks are common in federated learning environments, manipulating the federated learning model to bypass the global model. The proposed work resists model poisoning attacks by adopting homomorphic encryption, which successfully prevents such attacks.
- **Cryptographic Attacks:** In conventional IoMT environments, systems are highly vulnerable to cryptographic attacks, where conventional cryptographic methods are easily spoofed. The proposed work addresses this issue by adopting a quantum-based authentication method that resists both conventional and recent cryptographic attacks to a greater extent.

D. Research Findings

The proposed method adopts Fed-IoMT-Block by utilizing blockchain and federated learning technologies. Further, the comparative results show that the proposed work outperforms other methods. The cost complexities of the proposed algorithms align with the multiple QAP systems in Figure 1, as depicted in Table VI.

VI. CONCLUSION

The poor security measures and heterogeneity issues in IoMT devices present challenges in the IoMT environment. The proposed Fed-IoMT-Block framework addresses these challenges by adopting a three-layered approach, consisting of an IoMT sensor layer, a cloud layer, and an application layer. The authenticity of IoMT clients is validated using the QAP protocol, based on several metrics such as ID and biometrics. Only authenticated IoMT device results are securely communicated using heterogeneous federated technology with homomorphic encryption through Att-CapsNet. Meanwhile, external network traffic is analyzed using the LGBM algorithm based on several metrics.

The implemented algorithm classifies traffic into normal and malicious categories. Based on these classifications, a risk assessment is performed using various mathematical analyses. Finally, role-based access control is implemented based on the roles registered during the registration phase in the application layer. This role-based access control mechanism prevents malicious unauthorized access and ensures end-to-end security. The performance results demonstrate that the proposed model outperforms existing models. This work will be further enhanced by introducing a stronger encryption algorithm, protection mechanisms against potential cyber-attacks, energy consumption considerations, and real-world testing.

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